

Questions

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Q1 - 2024 (04 Apr Shift 1)

The electric field in an electromagnetic wave is given by $\vec{E} = \hat{i}40 \cos \omega(t - z/c) \text{ NC}^{-1}$. The magnetic field induction of this wave is (in SI unit) :

(1) $\vec{B} = \hat{k} \frac{40}{c} \cos \omega(t - z/c)$

(2) $\vec{B} = \hat{j}40 \cos \omega(t - z/c)$

(3) $\vec{B} = \hat{i} \frac{40}{c} \cos \omega(t - z/c)$

(4) $\vec{B} = \hat{j} \frac{40}{c} \cos \omega(t - z/c)$

Q2 - 2024 (04 Apr Shift 2)

Arrange the following in the ascending order of wavelength:

A. Gamma rays (λ_1)

B. x - rays (λ_2)

C. Infrared waves (λ_3)

D. Microwaves (λ_4)

Choose the most appropriate answer from the options given below

(1) $\lambda_4 < \lambda_3 < \lambda_1 < \lambda_2$

(2) $\lambda_2 < \lambda_1 < \lambda_4 < \lambda_3$

(3) $\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$

(4) $\lambda_4 < \lambda_3 < \lambda_2 < \lambda_1$

Q3 - 2024 (05 Apr Shift 1)

An alternating voltage of amplitude 40 V and frequency 4kHz is applied directly across the capacitor of $12\mu\text{F}$.

The maximum displacement current between the plates of the capacitor is nearly :

(1) 10 A

(2) 12 A

(3) 8 A

(4) 13 A

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Q4 - 2024 (05 Apr Shift 2)

Match List-I with List-II :

	List-I		List-II
	EM-Wave		Wavelength Range
(A)	Infra-red	(I)	$< 10^{-3}$ nm
(B)	Ultraviolet	(II)	400 nm to 1 nm
(C)	X-rays S	(III)	1 mm to 700 nm
(D)	Gamma rays	(IV)	1 nm to 10^{-3} nm

(1) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

(2) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(4) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

Q5 - 2024 (06 Apr Shift 1)

Electromagnetic waves travel in a medium with speed of $1.5 \times 10^8 \text{ m s}^{-1}$. The relative permeability of the medium is 2.0 . The relative permittivity will be:

(1) 2

(2) 4

(3) 5

(4) 1

Q6 - 2024 (06 Apr Shift 2)

In the given electromagnetic wave $E_y = 600 \sin(\omega t - kx) \text{ V m}^{-1}$, intensity of the associated light beam is (in W/m^2 : (Given $\epsilon_0 = 9 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$)

(1) 243

(2) 729

(3) 972

(4) 486

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Q7 - 2024 (09 Apr Shift 1)

A plane EM wave is propagating along x direction. It has a wavelength of 4 mm. If electric field is in y direction with the maximum magnitude of 60 Vm^{-1} , the equation for magnetic field is :

(1) $B_z = 2 \times 10^{-7} \sin \left[\frac{\pi}{2} \times 10^3 (x - 3 \times 10^8 t) \right] \hat{k} \text{ T}$

(2) $B_z = 60 \sin \left[\frac{\pi}{2} (x - 3 \times 10^8 t) \right] \hat{k} \text{ T}$

(3) $B_x = 60 \sin \left[\frac{\pi}{2} (x - 3 \times 10^8 t) \right] \hat{i} \text{ T}$

(4) $B_z = 2 \times 10^{-7} \sin \left[\frac{\pi}{2} (x - 3 \times 10^8 t) \right] \hat{k} \text{ T}$

Q8 - 2024 (09 Apr Shift 2)

The magnetic field in a plane electromagnetic wave is $B_y = (3.5 \times 10^{-7}) \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) \text{ T}$. The corresponding electric field will be :

(1) $E_y = 10.5 \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) \text{ Vm}^{-1}$

(2) $E_z = 1.17 \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) \text{ Vm}^{-1}$

(3) $E_y = 1.17 \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) \text{ Vm}^{-1}$

(4) $E_z = 105 \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) \text{ Vm}^{-1}$

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Answer Key

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Q1 (4) [//. mathongo](#) **Q2** (3) [//. mathongo](#) **Q3** (2) [//. mathongo](#) **Q4** (1) [//. mathongo](#)

Q5 (1) [//. mathongo](#) **Q6** (4) [//. mathongo](#) **Q7** (1) [//. mathongo](#) **Q8** (4) [//. mathongo](#)

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Solutions

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Q1

$$\vec{E} = \hat{i} 40 \cos \omega \left(t - \frac{z}{c} \right)$$

$\vec{E}$ is along +x direction

$\vec{v}$ is along +z direction

So direction of $\vec{B}$ will be along +y and magnitude of B will be $\frac{E}{c}$

So answer is $\frac{40}{c} \cos \omega \left(t - \frac{z}{c} \right) \hat{j}$

Q2

$$\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$$

Q3

Displacement current is same as conduction current in capacitor.

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

$$= \frac{1}{2\pi \times 4 \times 10^3 \times 12 \times 10^{-6}} = 3.317 \Omega$$

$$I = \frac{V}{X_C} = \frac{40}{3.317} = 12 \text{ A}$$

Q4

Infrared is the least energetic thus having biggest wavelength (λ) & gamma rays are most energetic thus having smallest wavelength (λ).

Q5

$$\frac{\epsilon_m \times \mu_m}{\epsilon_0 \times \mu_0} = \frac{1}{v^2}$$

$$\epsilon_r \times \mu_r = \frac{c^2}{v^2}$$

$$\epsilon_r \times 2 = \frac{(3 \times 10^8)^2}{(1.5 \times 10^8)^2}$$

$$\epsilon_r \times 2 = 4$$

$$\epsilon_r = 2$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

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Solutions

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Q6

$$\begin{aligned}
 \text{Intensity} &= \frac{1}{2} \epsilon_0 E_0^2 c \\
 &= \frac{1}{2} \times 9 \times 10^{-12} \times (600)^2 \times 3 \times 10^8 \\
 &= \frac{9}{2} \times 36 \times 3 = 486 \text{ W/m}^2
 \end{aligned}$$

Q7

$$E = BC \Rightarrow 60 = B \times 3 \times 10^8$$

$$\Rightarrow B = 2 \times 10^{-7}$$

$$\text{Also } C = f\lambda$$

$$\Rightarrow 3 \times 10^8 = f \times 4 \times 10^{-3}$$

$$\Rightarrow f = \frac{3}{4} \times 10^{11}$$

$$\Rightarrow \omega = 2\pi f = \frac{3}{4} \times 2\pi \times 10^{11}$$

$$\Rightarrow \omega = \frac{\pi}{2} \times 10^{11} \text{ rad/s}$$

$$\Rightarrow \text{Electric field} \Rightarrow y \text{ direction}$$

$$\text{Propagation} \Rightarrow x \text{ direction}$$

$$\text{Magnetic field} \Rightarrow z\text{-direction}$$

Q8

$$E_0 = B_0 C$$

$$E_0 = 3 \times 10^8 \times (3.5 \times 10^{-7}) \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t)$$

$$E_0 = 105 \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) \text{ Vm}^{-1}$$

Data inconsistent while calculating speed of wave. You can challenge for data.